

Introduction to machine learning

Linear Regression

Introduction to machine learning

Linear Regression Models -

- a. The term "regression" generally refers to predicting a real number. However, it can also be used for classification (predicting a category or class.)
- b. The term "linear" in the name "linear regression" refers to the fact that the method models data with linear combination of the explanatory variables.
- c. A linear combination is an expression where one or more variables are scaled by a constant factor and added together.
- d. In the case of linear regression with a single explanatory variable, the linear combination used in linear regression can be expressed as:

$$\text{response} = \text{intercept} + \text{constant} * \text{explanatory}$$

- e. In its most basic form fits a straight line to the response variable. The model is designed to fit a line that minimizes the squared differences (also called errors or residuals.).

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Linear Regression Models -

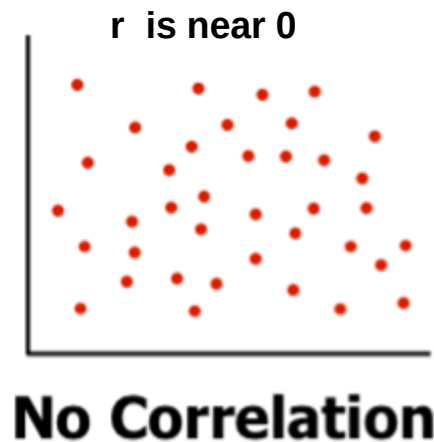
- a. Before we generate a model, we need to understand the degree of relationship between the attributes Y and X
- b. Mathematically correlation between two variables indicates how closely their relationship follows a straight line. By default we use Pearson's correlation which ranges between -1 and +1.
- c. Correlation of extreme possible values of -1 and +1 indicate a perfectly linear relationship between X and Y whereas a correlation of 0 indicates absence of linear relationship
 - I. When r value is small, one needs to test whether it is statistically significant or not to believe that there is correlation or not

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Linear Regression Models -

- d. Coefficient of relation - Pearson's coefficient $p(x,y) = \text{Cov}(x,y) / (\text{std Dev}(x) \times \text{std Dev}(y))$

$$r_{xy} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2} \sqrt{\sum (y_i - \bar{y})^2}}$$



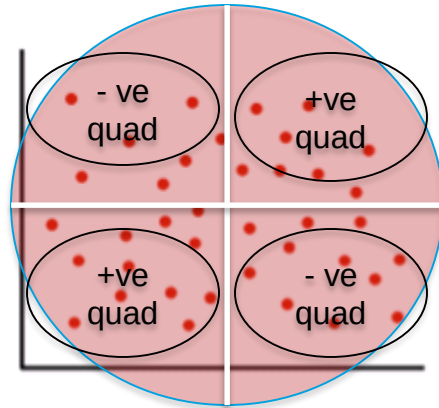
- e. **Generating linear model for cases where r is near 0**, makes no sense. The model will not be reliable. For a given value of X, there can be many values of Y! Nonlinear models may be better in such cases

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Linear Regression Models (Recap) -

- f. Coefficient of relation - Pearson's coefficient $p(x,y) = \text{Cov}(x,y) / (\text{std Dev } (x) \times \text{std Dev } (y))$

$$r_{xy} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2} \sqrt{\sum (y_i - \bar{y})^2}}$$



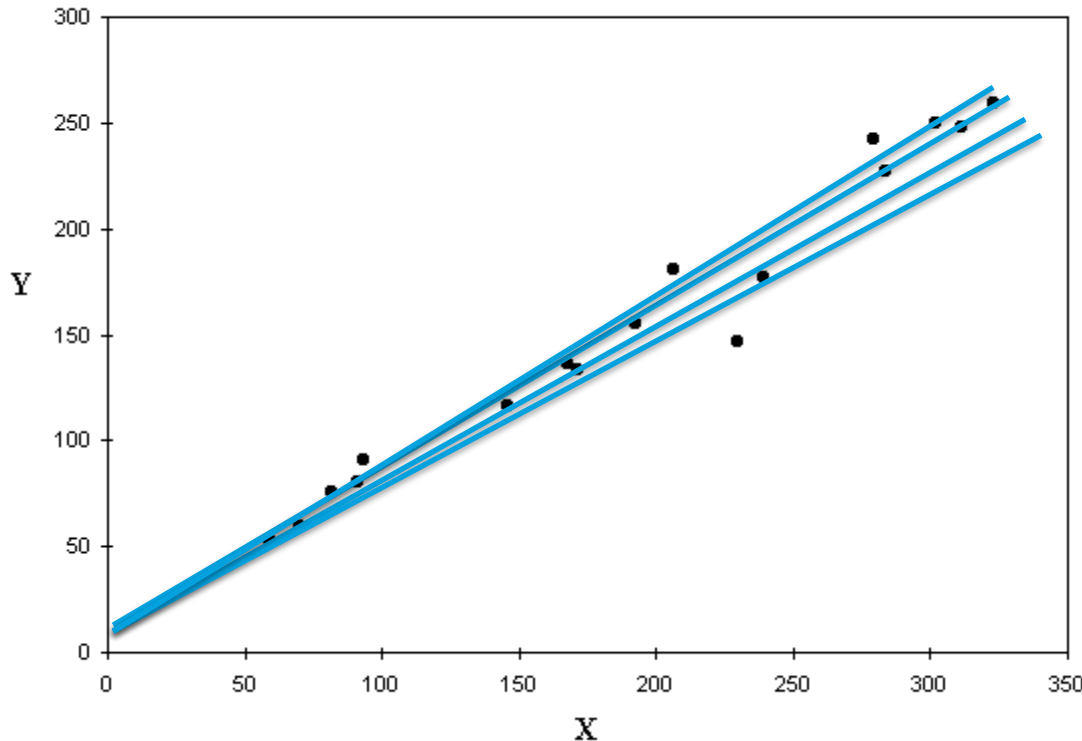
$$r_{xy} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2} \sqrt{\sum (y_i - \bar{y})^2}} = 0$$

<http://www.socscistatistics.com/tests/pearson/Default2.aspx>

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Linear Regression Models

- g. Given $Y = f(x)$ and the scatter plot shows apparent correlation between X and Y
Let's fit a line into the scatter which shall be our model
- h. But there are infinite number of lines that can be fit in the scatter. Which one should we consider as the model?

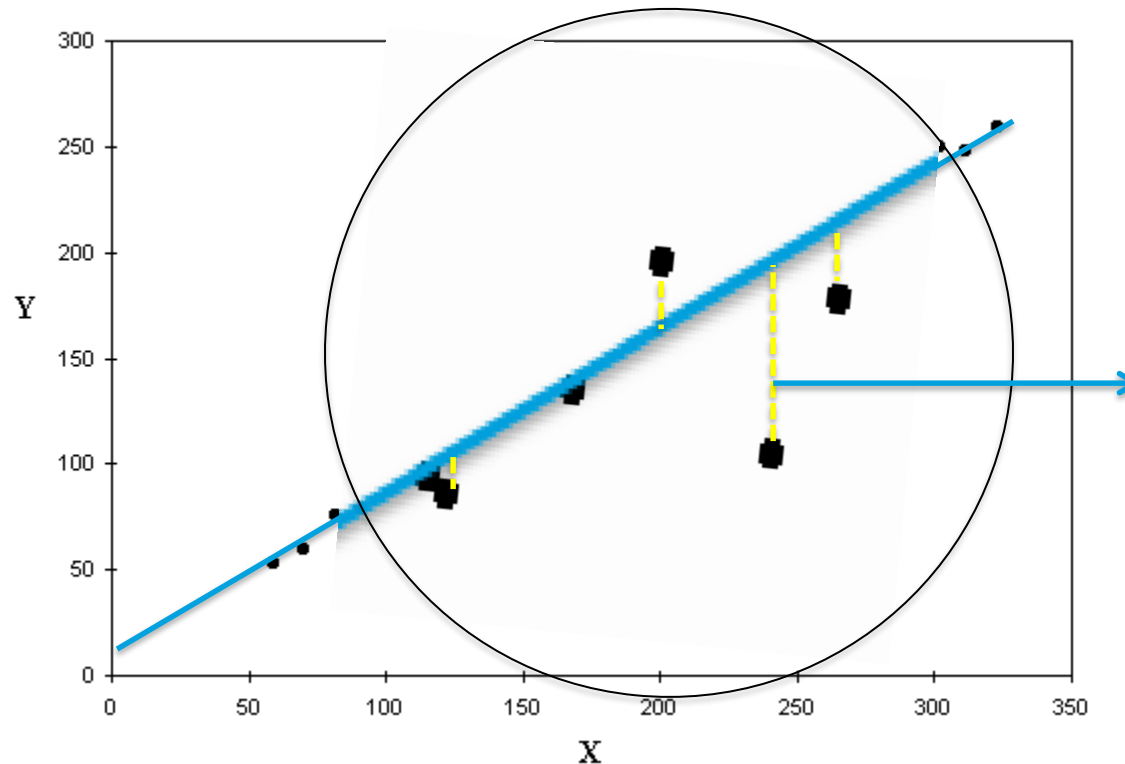


- i. This and many other algorithms use gradient descent or variants of gradient descent method for finding the best model
- j. Gradient descent methods use partial derivatives on the parameters (slope and intercept) to minimize sum of squared errors

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Linear Regression Models (Recap) -

- k. Whichever line we consider as the model, it will not pass through all the points.
- l. The distance between a point and the line (drop a line vertically (shown in yellow)) is the error in prediction
- m. That line which gives least sum of squared errors is considered as the best line



$$\text{Error} = (T - (mx + C))$$

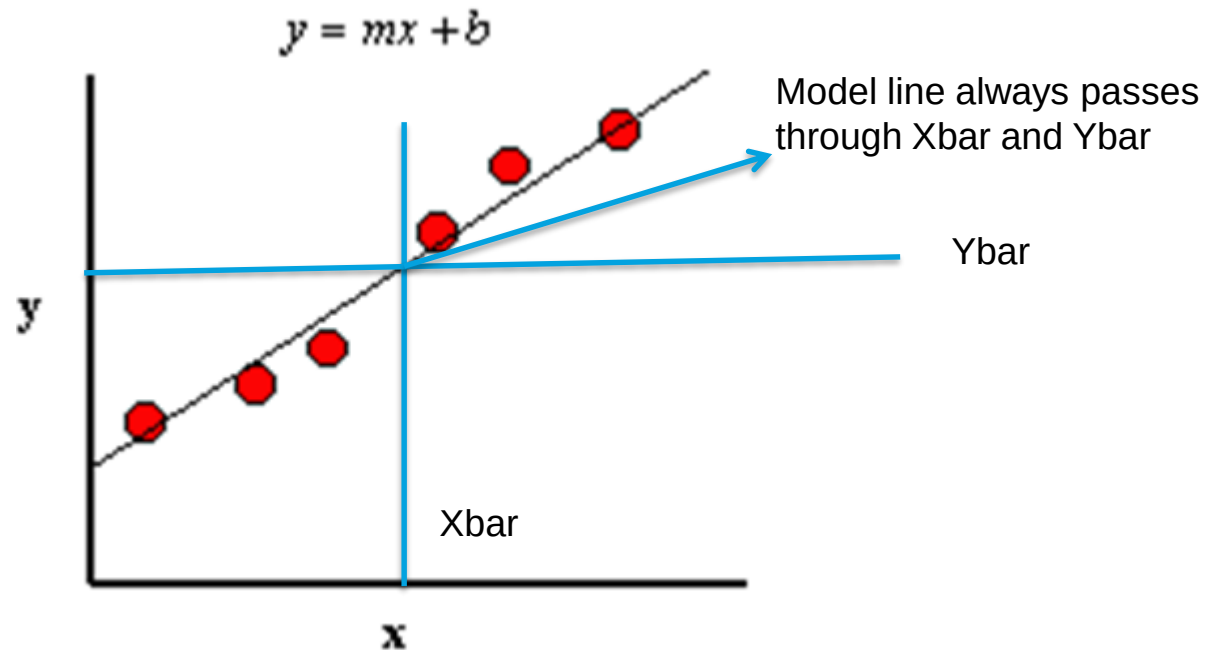
Sum of all errors can cancel out and give 0

We square all the errors and sum it up. That line which gives us least sum of squared errors is the best fit

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Linear Regression Models -

- n. Coefficient of determinant – determines the fitness of a linear model. The closer the points get to the line, the R^2 (coeff of determinant) tends to 1, the better the model is

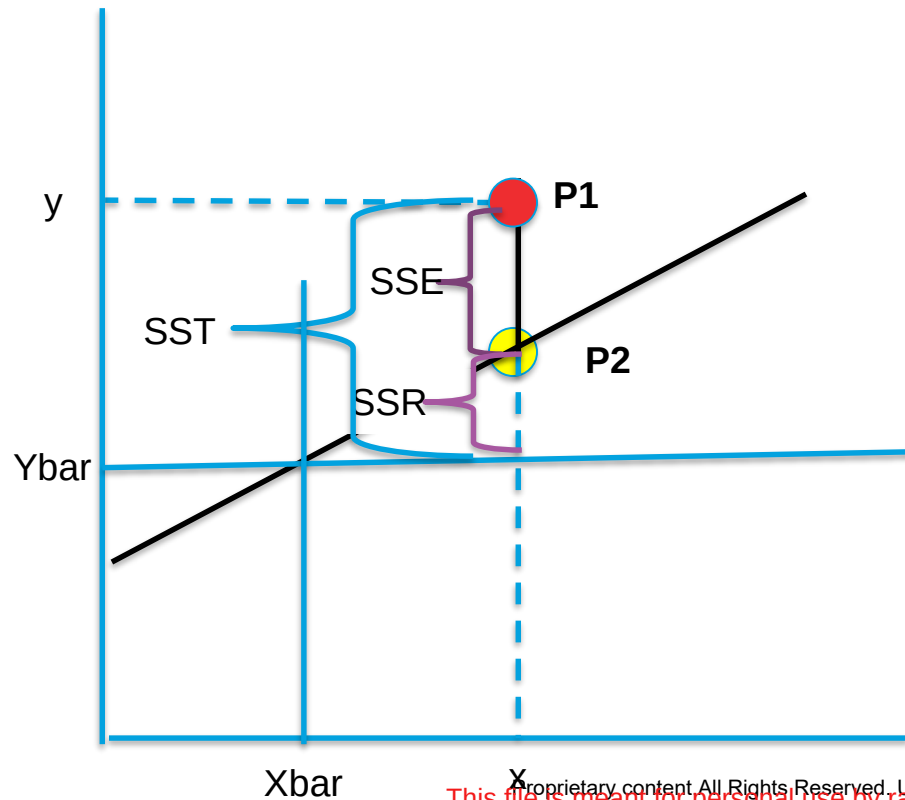


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Linear Regression Models -

o. Coefficient of determinant (Contd...)

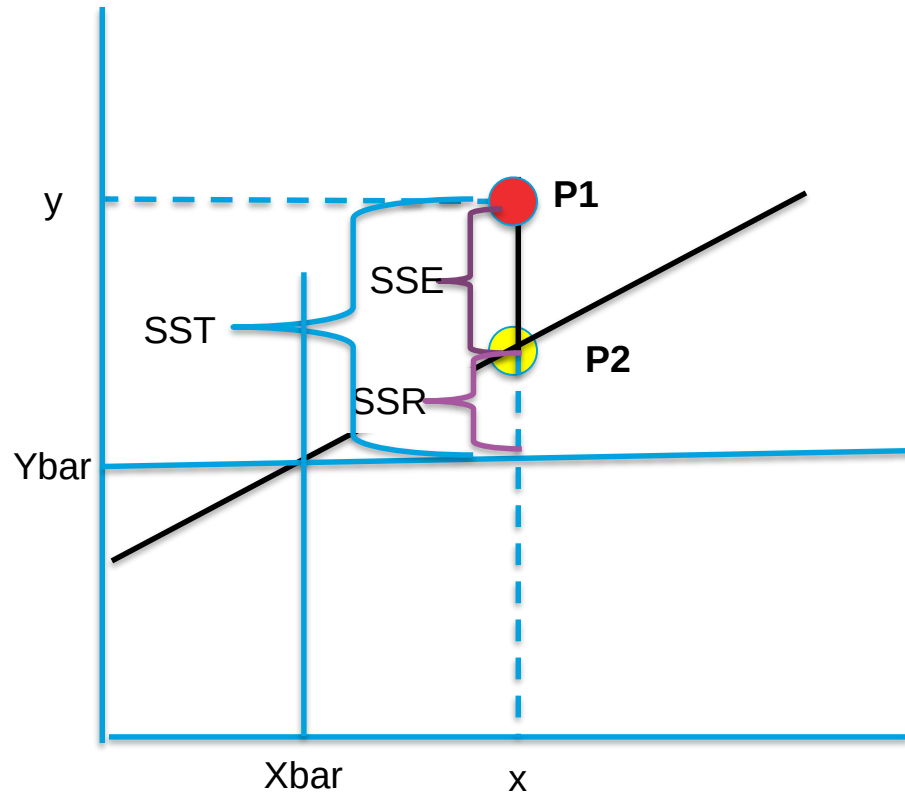
- I. There are a variety of errors for all those points that don't fall exactly on the line.
- II. It is important to understand these errors to judge the goodness of fit of the model i.e. How representative the model is likely to be in general
- III. Let us look at point P1 which is one of the given data points and associated errors due to the model



1. P1 – Original y data point for given x
2. P2 - Estimated y value for given x
3. Ybar – Average of all Y values in data set
4. SST – Sum of Square error Total (SST)
Variance of P1 from Ybar $(Y - Ybar)^2$
5. SSR - Regression error $(p2 - ybar)^2$ (portion SST captured by regression model)
6. SSE - Residual error $(p1 - p2)^2$

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Linear Regression Models -



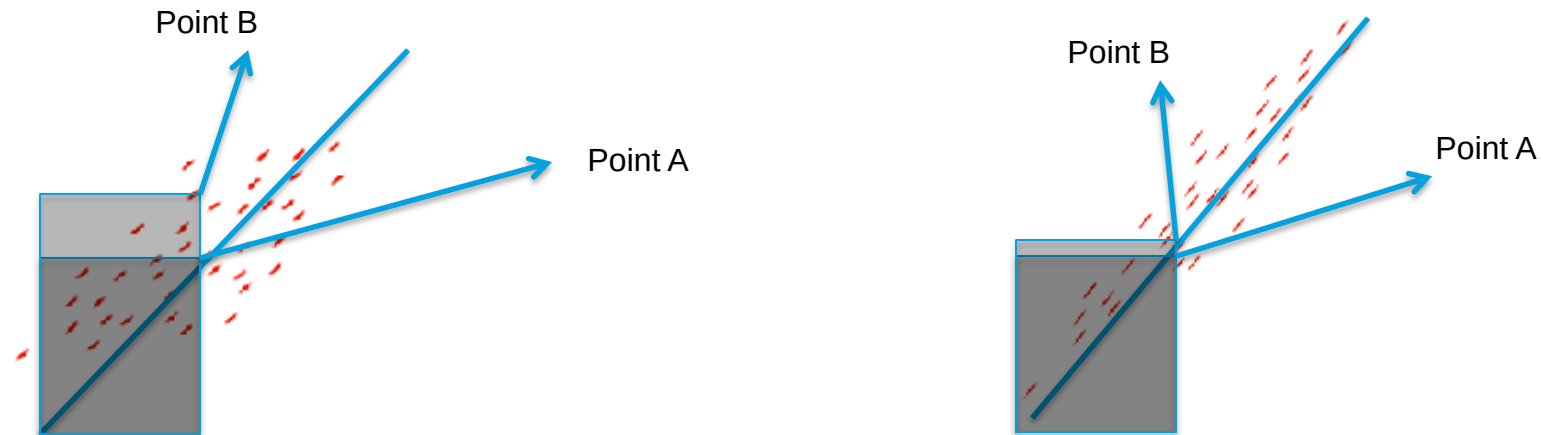
p. Coefficient of determinant (Contd...)

1. That model is the most fit where every data point lies on the line. i.e. $SSE = 0$ for all data points
2. Hence SSR should be equal to SST i.e. SSR/SST should be 1.
3. Poor fit will mean large SSE . SSR/SST will be close to 0
4. SSR / SST is called as r^2 (r square) or coefficient of determination
5. r^2 is always between 0 and 1 and is a measure of utility of the regression model

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Linear Regression Models -

q. Coefficient of determinant (Contd...) -



In case of point “A”, the line explains the variance of the point

Whereas point “B” the is a small area (light grey) which the line does not represent.

%age of total variance that is represented by the line is coeff of determinant

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Linear Regression Model -

Advantages –

1. Simple to implement and easier to interpret the outputs coefficients

Disadvantages -

1. Assumes a linear relationships between dependent and independent variables. That is, it assumes there is a straight-line relationship between them
2. Outliers can have huge effects on the regression
3. Linear regression assume independence between attributes
4. Linear regression looks at a relationship between the mean of the dependent variable and the independent variables.
5. Just as the mean is not a complete description of a single variable, linear regression is not a complete description of relationships among variables
6. Boundaries are linear

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Linear Regression Model -

Lab- 1- Estimating mileage based on features of a second hand car

Description – Sample data is available at

<https://archive.ics.uci.edu/ml/datasets/Auto+MPG>

The dataset has 9 attributes listed below that define the quality

1. mpg: continuous
2. cylinders: multi-valued discrete
3. displacement: continuous
4. horsepower: continuous
5. weight: continuous
6. acceleration: continuous
7. model year: multi-valued discrete
8. origin: multi-valued discrete
9. car name: string (unique for each instance)

Sol : mpg-linear regression.ipynb

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